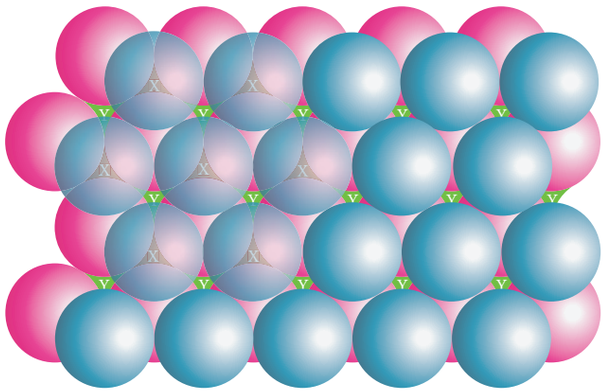
Wherever a sphere of second layer (b) is above the void (x) of the first layer (a), a tetrahedral void is formed. This constitutes four spheres – three in the lower (a) and one in the upper layer (b). When the centers of these four spheres are joined, a tetrahedron is formed.

At the same time, the voids (y) in the first layer (a) are partially covered by the spheres of layer (b), now such a void in (a) is called a octahedral void. This constitutes six spheres – three in the lower layer (a) and three in the upper layer (b). When the centers of these six spheres are joined, an octahedron is formed. Simultaneously new tetrahedral voids (or holes) are also created by three spheres in second layer (b) and one sphere of first layer (a)



The number of voids depends on the number of close packed spheres. If the number of close packed spheres be 'n' then, the number of octahedral voids generated is equal to n and the number of tetrahedral voids generated is equal to 2n.

Formation of third layer:

The third layer of spheres can be formed in two ways to achieve closest packing

(i) aba arrangement - hcp structure

(ii) abc arrangement – ccp structure

The spheres can be arranged so as to fit into the depression in such a way that the third layer is directly over a first layer as shown in the figure. This "aba" arrangement is known as the hexagonal close packed (hcp) arrangement. In this arrangement, the tetrahedral voids of the second layer are covered by the spheres of the third layer.

Alternatively, the third layer may be placed over the second layer in such a way that all the spheres of the third layer fit in octahedral voids. This arrangement of the third layer is different from other two layers (a) and (b), and hence, the third layer is designated (c). If the stacking of layers is continued in abcabcabc… pattern, then the arrangement is called cubic close packed (ccp) structure.

In both hcp and ccp arrangements, the coordination number of each sphere is 12 – six neighbouring spheres in its own layer, three spheres in the layer above and three sphere in the layer below. This is the most efficient packing.

XII U6 Solid State - Jerald.indd 190 04-01-2021 13:28:11